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VAN / SURNAME: _____

VOORLETTERS / INITIALS: _____ REG.No.: _____

HANDTEKENING / SIGNATURE: _____

KURSUS / COURSE: FSK 116

UNIVERSITEIT VAN PRETORIA
THE UNIVERSITY OF PRETORIA

FSK 116

EKSAMEN / EXAMINATION

29 MEI / MAY 2000

Totale Punte / Total Marks: 100

Tyd / Time: 3 ure / hours

- Beantwoord alle vrae. / Answer all questions.
- Programmeerbare sakrekenaars mag NIE gebruik word NIE
- Programmable pocket calculators may NOT be used.
- Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.
All test and exam regulations of the University of Pretoria are applicable.

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FSK 116: EKSAMEN / EXAMINATION
29 MEI / MAY 2000

VAN EN VOORLETTERS: _____
SURNAME AND INITIALS:

STUDENTE NR. _____
STUDENT NUMBER:

HANDTEKENING: _____
SIGNATURE:

- (c) Bepaal die hoogte h_2 van m_2 na die botsing.
To what height h_2 does 2 swing after the collision?

(3)

VRAAG / QUESTION 3

- (a) Definieer hoeksnelheid ω en gee die eenhede daarvan.
Define angular velocity ω and give its units.

(2)

- (b) (i) 'n Slypwiël begin draai (vanuit rus) met 'n konstante hoekversnelling $\alpha = 0,35 \text{ rad/s}^2$.
 Bepaal die hoekverplasing θ wat die wiel in 18 s maak.
A grindstone starts from rest and has a constant angular acceleration $\alpha = 0,35 \text{ rad/s}^2$. What is the angular displacement θ at $t = 18 \text{ s}$?

(3)

- (ii) Hoeveel omwentelinge het die wiel gemaak in 18 s?
Determine the number of revolutions completed in the 18 s.

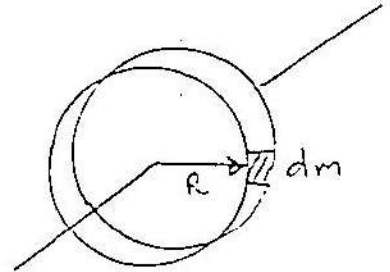
(1)

- (iii) Wat is die wiel se hoeksnelheid na 18 s?
What is the wheel's angular velocity at $t = 18 \text{ s}$?

(2)

- (c) 'n Dun ring met massa m en straal R roteer om 'n as deur sy middelpunt. Toon aan dat die traagheidsmoment van die hele ring is $I = mR^2$.

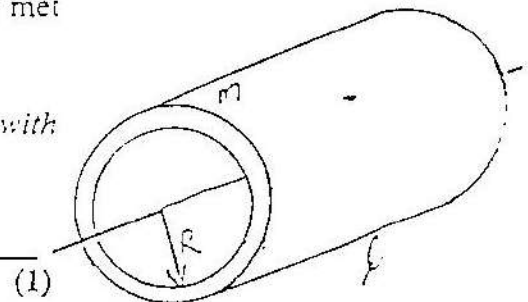
A thin ring has a mass m and radius R and it rotates about an axis through its center. Show that the moment of inertia of the whole ring is $I = mR^2$.



(2)

- (d) Wat is die traagheidsmoment van 'n dun silinder met straal R , massa m en lengte ℓ ?

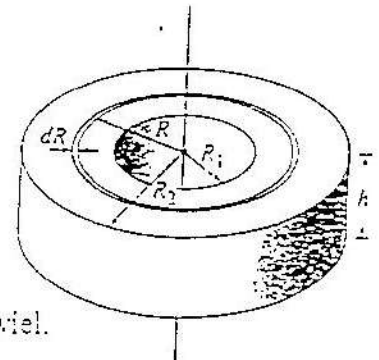
What is the moment of inertia of a thin cylinder with radius R , mass m and length ℓ ?



(1)

- (e) 'n Slypwiël het 'n gat in die middel met 'n binne radius R_1 , 'n buiteradius R_2 en 'n massa M .

A grindstone with a hole in the center has an inner radius R_1 , an outer radius R_2 and a mass M .



- (i) Skryf 'n uitdrukking neer vir die volume V van die wiel.
Give an expression for the volume V of the wheel.

(1)

- (ii) Beskou die wiel as opgebou uit dun ringe met dikte dR . Gee 'n uitdrukking vir die volume dV van 'n ring met straal R , dikte dR en hoogte h .
The cylinder can be constructed from a number of thin concentric cylindrical rings of thickness dR . Give an expression for the volume dV of a ring with radius R , thickness dR and height h .

(1)

- (iii) 'n Dun ring met massa dm , volume dV en dikte dR se digtheid is ρ . Gee 'n uitdrukking vir dm in terme van ρ .

The mass of a thin ring is dm , its volume is dV and its thickness is dR . The ring has a density ρ . Give an expression for dm in terms of ρ .

- (iv) Met bostaande uitdrukking vir dm , herlei 'n uitdrukking vir die traagheidsmoment I vir die wiel in terme van R_1 en R_2 .

With the expression for dm above, derive an expression for the moment of inertia I for the whole wheel in terms of R_1 and R_2 .

- (v) Indien die totale massa van die wiel M is en sy digtheid is ρ , skryf 'n uitdrukking vir ρ in terme van M , R_1 en R_2 .

The wheel has a total mass M and density ρ . Write an expression for ρ in terms of M , R_1 and R_2 .

- (vi) Met $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, toon aan dat:

$$I = \frac{1}{2} M (R_2^2 + R_1^2).$$

With $R_2^4 - R_1^4 = (R_2^2 - R_1^2)(R_2^2 + R_1^2)$, show that

$$I = \frac{1}{2} M (R_2^2 + R_1^2).$$

- (vii) Die wiel het 'n massa $M = 5 \text{ kg}$ met $R_1 = 0,1 \text{ m}$ en $R_2 = 0,5 \text{ m}$. Bepaal die waarde van I .

If the wheel has a mass $M = 5 \text{ kg}$, with $R_1 = 0,1 \text{ m}$ and $R_2 = 0,5 \text{ m}$, determine the value of I .

(2)

- (viii) Die wiel roteer met 'n hoeksnelheid $\omega = 1000$ omwentelings per minuut. Bepaal die hoeksnelheid in rad/s .

The wheel is rotating with an angular velocity $\omega = 1000$ revolutions per minute. Determine the angular velocity in rad/s .

(2)

- (ix) Bepaal die kinetiese energie wat in die wiel gestoor is.

What is the kinetic energy stored in the wheel?

(2)

- (x) Indien al die rotasie kinetiese energie oorgedra word aan 'n 60 W gloeilamp, hoe lank sal die gloeilamp brand?

If all the rotational kinetic energy is transferred to a 60 W globe, for how long will the lamp light up?

(3)

VRAAG / QUESTION 4

- (a) 'n Veer word oor 'n afstand x saamgedruk. Gee Hooke se Wet vir die krag van die veer.
A spring is compressed over a distance x . Give Hooke's law for the spring force.

(1)

- (b) Gee die formule om die arbeid W te bepaal wanneer 'n konstante krag $\bar{F}$ oor 'n afstand x verplaas.

Give a formula for the work W done when a force $\bar{F}$ acts over a distance x .

(1)

- (c) Gee 'n formule vir die arbeid W wat verrig word deur 'n krag $\bar{F}(x)$, wat nie konstant is nie, tussen die punte x_i en x_f .

Give a formula for the work W done by a force $\bar{F}(x)$, which is not constant, between two points x_i and x_f .

(1)

- (d) Toon aan dat die arbeid wat verrig word deur die krag $\bar{F}(x)$ van 'n veer gegee word deur $W = -\frac{1}{2} kx^2$ indien $x_i = 0$ en $x_f = x$.

Show that the work done by a spring force $\bar{F}(x)$ is given by $W = -\frac{1}{2} kx^2$ if $x_i = 0$ and $x_f = x$.

(3)

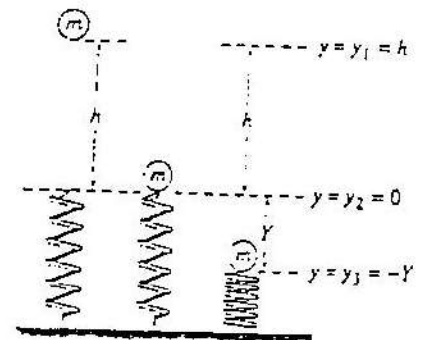
- (e) Indien die verandering in potensiële energie $\Delta U = -W$, wat is die potensiële energie wat gestoor is in 'n veer wat 'n afstand x saamgedruk is?

If the change in potential energy $\Delta U = -W$, what is the potential energy stored in a spring which is compressed a distance x .

(1)

- (f) 'n Bal met massa $m = 2,60$ kg val vertikaal afwaarts vanuit rus deur 'n hoogte $h = 55$ cm. Dit val op 'n veer wat dan $Y = 15$ cm saamdruk.

A ball with mass $m = 2,60$ kg, starting from rest, falls a vertical distance $h = 55$ cm before striking a coiled spring, which compresses an amount $Y = 15$ cm.



- (i) Toon met behoud van energie aan dat die snelheid v_2 van die bal net wanneer dit die veer tref (by posisie $y_2 = 0$) is $3,28$ m/s.

From the conservation of energy show that the velocity v_2 of the ball just as it touches the spring (at position $y_2 = 0$) is $3,28$ m/s.

(4)

- (ii) Bepaal die veerkonstante k indien die veer ten volle saamgedruk word tot by $y_3 = -Y = -0,15$ m

Determine the spring constant k if the spring is fully compressed at:

$y_3 = -Y = -0,15$ m

(5)

- (g) Met veerkonstante $k = 1583 \text{ N/m}$, bepaal die posisie(s) y_4 (relatief tot y_2) waar $v_4 = 3 \text{ m/s}$. Indien u meer as een antwoord vir y_4 kry, verduidelik kortliks hul betekenis.
- If the spring constant $k = 1583 \text{ N/m}$, determine the position(s) y_4 (relative to y_2) where $v_4 = 3 \text{ m/s}$. If you find more than one answer for y_4 , give a short description of their meaning.*

(6)

VRAAG/QUESTION 5

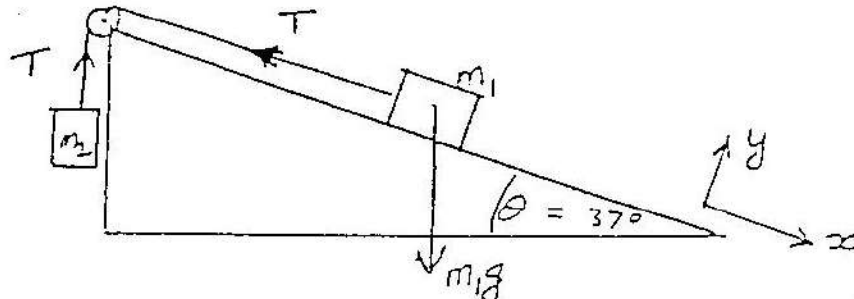
- (a) Kragte wat op 'n liggaam inwerk is sodanig da: $\sum \vec{F} \neq 0$. Gee Newton se tweede wet.
For different forces acting on an object $\sum \vec{F} \neq 0$. Give Newton's second law.

(1)

- (b) Gee 'n formule vir die statiese wrywingskrag f_s , indien die wrywingskoëffisiënt μ_s is.
Give a formula for the static frictional force f_s with the static coefficient of friction μ_s .

(2)

- (c) Beskou die volgende opstelling:
Consider the following setup:



Indien die statiese wrywingskoëffisiënt $\mu_s = 0,4$ is moet ons die waardes van m_2 bepaal wat die sisteem in rus sal hou.

If the coefficient of static friction is $\mu_s = 0,4$ we have to determine the range of values for m_2 to keep the system at rest.

- (i) As m_2 klein is, sal m_1 neig om af te gly. / With m_2 small, m_1 would tend to slide down.

Die som van die kragte in y-rigting op m_1 is : / The sum of the forces in y-direction on m_1 is:

_____ (1)

Die som van die kragte in x-rigting op m_1 is : / The sum of the forces in x-direction on m_1 is:

_____ (1)

- (ii) Spankrag T in die tou weens m_2 : / Tension in the rope due to m_2 :

_____ (1)

- (iii) Die normaalkrag N op m_1 is: / The normal force N on m_1 is:

_____ (1)

- (iv) Met $m_1 = 10$ kg, toon aan dat $m_2 = 2,8$ kg moet wees. / With $m_1 = 10$ kg, show that $m_2 = 2,8$ kg.

_____ (4)

- (v) As m_2 groot is, sal m_1 neig om op te beweeg.
With m_2 large, m_1 would tend to move upwards.

Die som van die kragte in die x-rigting op m_1 (gee in simbole): / *The sum of the forces in the x-direction on m_1 (give in symbols):*

- (vi) Bepaal nou m_2 met $m_1 = 10$ kg. / *Now, determine the value of m_2 with $m_1 = 10$ kg* (4)

- (vii) As $m_2 = m_1 = 10$ kg. en $\mu_k = 0,3$ is, toon aan die versnelling van die sisteem is $a = 0,78 \text{ m/s}^2$.
 With $m_2 = m_1 = 10$ kg. and $\mu_k = 0,3$, show that the acceleration of the system is $a = 0,78 \text{ m/s}^2$. (4)